

StudyReel — Card Catalog Pivot: Handoff Spec

Design direction for opencode integration

September 2026

0. What this document is

This is the full spec for replacing StudyReel’s Instagram-shaped feed (avatar, likes, follow, comments, mastery ring, hashtag pills) with a **card catalog** metaphor: shelves, a stack of index cards, three filing directions, and a cabinet of drawers. It covers the interaction spec, the data model changes, the API surface, the design tokens, a migration map from the old fields, and the full source of the three mockups you already have as HTML files.

Everything here is additive to your Phase 1/2 work (ingestion, Gemini generation, render pipeline, Publisher abstraction). **Nothing about ingestion, rendering, or the Publisher seam changes.** Only the feed/consumption layer and its data model change.

1. Why the pivot (one paragraph, for your report)

The original feed reused Instagram’s grammar — avatar, verified check, follower-style ring, heart, hashtags, single-axis swipe — because that’s the fastest way to prototype “a feed.” But every one of those elements imports Instagram’s *incentive structure* (perform for followers, chase likes) into a study tool, which fights the actual goal (retention of material, not attention). The card catalog metaphor borrows from something students already do — sorting physical flashcards — and gives every gesture a study-specific meaning instead of a social one.

2. Interaction spec

2.1 Screen: Deck (the primary swipe screen)

Layout, top to bottom:

1. **Header** — wordmark “StudyReel” + tagline “a card catalog, not a feed.” Light/dark toggle, top right. No live/pulse indicator (that was a “someone’s online” signal borrowed from stories — not needed here).
2. **Shelf row** — horizontally scrollable row of **spines**, one per module/tag (VTU modules by name, or global tags AI/EV/Aero/CS). Each spine is a small vertical bar:
 - Fill height = mastery % for that shelf (see §3.2 for the calculation — **mastered cards only**, not catalog or review).
 - Tapping a spine swaps the active deck and resets the stack to that shelf’s unfiled cards.
 - Active spine gets a cyan outline. No numeric label on the spine itself (percentage fill only, no printed “52%” — visual only, per your last note to stop narrating mechanics).

3. **Stack** — the current shelf’s unfiled cards, rendered as a physical stack: top card at full opacity/scale, two more peeking behind at reduced scale/opacity. When a card is filed, the next one slides up to take its place.
4. **Card** — an index card:
 - Top-left: a small punch-hole (decorative, reinforces “real card” read).
 - Header row: module/tag chip (monospace) + read-time chip (“30s”).
 - Body: header (≤ 30 chars), body copy (≤ 140 chars), optional syntax-highlighted code block.
 - Footer: exam-relevance label + “flip card” control.
 - **Flip** (tap, not swipe) shows the back: exam-specific context, a harder follow-up, or the “why this matters” layer. This is a new content field per microtopic (see §3.1).
5. **Trays row** — three static tray labels (Review / Catalog / Mastered), purely as filing destinations. No counts shown.
6. **Filing controls** — three buttons (or, on real mobile, drag gestures) mapped to three directions:
 - **Left** → **Review** (amber). “I don’t know this yet.”
 - **Down** → **Catalog** (cyan). “Keep this, no pressure to know it.” This is the tray for curiosity-driven global users.
 - **Right** → **Mastered** (green). “I know this.”
 - No on-screen text explains what happens when you file a card — the motion (card flies in that direction into that tray) is the only feedback. This is deliberate (see §6, open item on onboarding).

What was removed vs. the old feed spec, explicitly, so nothing gets carried over by habit:

- Avatar / “studyreel.ai” identity block — gone. There is no account posting to you.
- Live pulsing dot — gone.
- Mastery ring around an avatar — replaced by shelf spine fill (§3.2).
- Heart / “Helpful” like counter, double-tap-to-like — gone. Filing a card *is* the reaction.
- Hashtag pills inside the caption — replaced by a single monospace tag chip per card, not tappable-as-filter from inside the card (filtering happens at the shelf level only).
- View count, eye icon — gone. There is no social proof to display.
- “Ask ?” affordance — deferred (this was Direction C in the earlier brainstorm; still future work, not in this pivot).
- Caption text block — folded into the card body; no separate “studyreel.ai wrote:” caption line.

2.2 Screen: My Cabinet (where filed cards live)

A cabinet with three **drawers**, one per tray (Review / Catalog / Mastered), each with a colored dot matching its filing color and a one-line description of what it’s for. Tapping a drawer’s face expands it (accordion — **only one drawer open at a time**, mimicking a real cabinet where you can’t sensibly have three drawers hanging open). Inside, each filed card renders as a compact row (tag + title) with a left-edge color stripe matching its tray color, tying this screen visually back to the Deck screen. Tapping a row opens that card’s full front/back view (reuses the Deck card component).

This is the destination for the “review queue” and “my catalog” that a purely swipe-based feed has no place to put.

3. Data model changes

3.1 New/changed fields on a microtopic (per-slide content)

Add a back field to whatever object currently holds header / body / code per slide — this is new content, not a reuse of the caption:

```
class MicroTopicSlide(BaseModel):
    header: str          # existing, <=30 chars
    body: str            # existing, <=140 chars
    code: str | None     # existing, optional, <=22 lines
    language: str | None # existing
    back_header: str     # NEW: <=30 chars, e.g. "Why it matters (exam)"
    back_body: str       # NEW: <=140 chars, exam angle / harder follow-up / context
    exam_weight: Literal["low", "medium", "high"] # NEW, replaces the old "□□□" free text
```

back_header/back_body should be generated by the same Gemini call that produces the front of the card (one extra field in the prompt/schema), then editable in the HITL review step alongside header/body/code — same live counters (≤ 30 / ≤ 140) apply.

3.2 New table: card filing state (per user, per slide)

This is the actual new state your product didn't have before — previously there was no concept of a user “doing” anything to a slide except viewing/liking it globally.

```
class CardStatus(str, Enum):
    unfiled = "unfiled"
    review = "review"
    catalog = "catalog"
    mastered = "mastered"

class UserCardState(BaseModel):
    user_id: str          # stub/anonymous id ok for prototype – no auth system required
    post_id: str
    slide_index: int
    status: CardStatus
    filed_at: datetime | None
    last_shown_at: datetime | None # for review re-surfacing, see 3.4
```

Prototype note: since the spec says “no per-user memory” was previously a constraint for the *global trending* loop, this table can still be scoped to a single local/anonymous user for the prototype — it does not require a real auth system, just a user_id you can hardcode or derive from a local device id. The global trending formula stays user-agnostic; only filing state is per-user.

3.3 Shelf fill calculation (replaces the mastery ring formula)

Old spec: mastery ring = “global saves/views %” (a vanity metric). New:

$\text{shelf_fill}(\text{shelf}) = \text{mastered_count}(\text{shelf}) / \text{total_slides}(\text{shelf})$

Do not include catalog or review in this number — mixing them back in makes the shelf fill meaningless (a curious browser who catalogs everything would look identical to a student who's mastered a module).

3.4 Review re-surfacing logic

- Cards with status = review should re-enter the shelf's stack on a later visit (simplest prototype version: re-inject them at the *end* of the current stack once the stack of unfiled cards is exhausted, rather than building real spaced-repetition scheduling — that's a "future work" line item, not required for the demo).
- Cards with status = catalog should **not** auto-resurface. They only appear in the Cabinet's Catalog drawer, browsed on the user's own terms.
- Cards with status = mastered also do not resurface by default.

3.5 Trending formula — optional small addition

Your existing formula: $\text{likes} * 0.7 + \text{views} * 0.15 - \text{hours} * 0.08$. Since likes/views no longer exist as UI concepts, replace with filing-derived signals if you want the global loop to react to catalog behavior:

$\text{trending_score} = \text{mastered_actions} * 0.6 + \text{catalog_actions} * 0.3 - \text{hours} * 0.08$

This is optional — if you're short on time, a simpler $\text{total_filings} * 0.7 - \text{hours} * 0.08$ is a fine prototype substitute and avoids inventing weights you can't defend in a viva.

4. API surface (additions to /api/v1 and /api/v2)

GET /api/v1/shelves
-> [{ shelf: "OS", total_slides: 24, mastered_count: 12, fill_pct: 0.50 }, ...]

GET /api/v1/deck?shelf=OS
-> unfiled + re-surfaced-review slides for that shelf, in stack order

POST /api/v1/file
body: { post_id, slide_index, status: "review"|"catalog"|"mastered" }
-> writes/updates UserCardState, recalculates shelf fill

GET /api/v1/cabinet?tray=review|catalog|mastered
-> filed cards for that tray, most-recently-filed first

These replace the old GET /api/feed?sort=trending|newest&tag=ai as the primary read path for the Deck screen. Keep the old trending/newest endpoint alive if you still want a global-discovery cold-start list feeding the initial shelf selection for a brand-new user — it's compatible with this model, it just no longer renders as a "feed."

5. Design tokens (unchanged from your original system, confirmed here for reference)

Token	Dark	Light
bg	#080c14	#f8fafc
surface	#0e1626	#ffffff
card	#131e33	#ffdf7
border	#1e2d4a	#e2e8f0

Token	Dark	Light
cyan (accent / mastered-adjacent... see below)	#38bdf8	#38bdf8
indigo	#6366f1	#6366f1
text	#e6edf7	#0f172a
muted	#8b9bb4	#64748b

New role colors introduced by filing (not in the original palette — add these):

Role	Hex	Used for
review (amber)	#fbbf24	review tray, review card stripe, spine hover state if you add one
catalog (cyan)	#38bdf8	catalog tray, catalog card stripe
mastered (green)	#34d399	mastered tray, mastered card stripe, shelf fill gradient top stop

Typography, radius, shadow, and motion tokens (Inter 400/600/700 + Fira Code, 16px card radius, 10px shadows, 0.35s cubic-bezier(0.2,0,0,1) transitions) are unchanged from your original design system doc.

6. Open items — decide before opencode starts wiring state

1. **First-run onboarding.** With no on-screen labels explaining the three filing directions, a first-time user needs *something* — a single one-time nudge on their very first card (“try filing this one”) is enough; don’t over-build this. Decide now whether it’s in scope for the demo or a named “future work” bullet in your report.
2. **Anonymous user id for the prototype.** Confirm how user_id is generated (local storage token, device id, or a single hardcoded demo user) since there’s no auth system in scope.
3. **Drag vs. buttons on real mobile.** The mockups use buttons for the three filing actions (reliable, demoable). If you want real drag-to-file gestures in the actual app, that’s a meaningfully bigger frontend lift (velocity/direction thresholds, snap-back on ambiguous drags) — decide if that’s worth the time versus buttons plus a swipe as a bonus, not a requirement.
4. **Global-curious vs. VTU framing on shared shelves.** AI/EV/Aero shelves are browsed by curious users who’ll lean on Catalog; VTU shelves are browsed by students under exam pressure who’ll lean on Review/Mastered. The UI is identical either way — worth deciding if that’s fine or if global shelves should default-hide the Mastered language (e.g., relabel Mastered as “Got it” only on non-VTU shelves) — cosmetic, not structural, so low risk to defer.

Appendix A — Deck screen, full source

See attached file `studyreel_card_catalog.html`.

Appendix B — My Cabinet screen, full source

See attached file `studyreel_my_cabinet.html`.

Appendix C — Earlier feed-shaped mockup (superseded)

See attached file `studyreel_feed_mockup.html`. Kept only for contrast/reference in your report (“here’s what we moved away from and why”) — do not build against this one.